

DAY-10 (12/11/2025)

Q) Write a program to find the given number is Prime or not a prime number?

Suppose without count variable.

```
num = int(input())
```

```
for i in range(2, num, 1):
```

```
    if (num % i == 0):
```

```
        print("not a prime number")
```

```
    else:
```

```
        print("prime number")
```

i/p $\Rightarrow$ 9

o/p $\Rightarrow$ prime number

not a prime number

prime number

prime number

prime number

prime number

prime number

Here we are getting for each iteration, because the print is inside if.

num % i (remainder)

1

2

3

4

5

6

7

8

1

0

1

4

3

2

1

Prime number
it is divisible by
1 & itself.

* Here every number
will be divisible by
1 & itself.

* So, we have to check
from 2, to the less than
of given number

like if it is 7

from 2, 3, 4, 5, 6 we
have to check.

o/p

prime number

not a prime number

prime number

prime number

prime number

prime number

prime number

with count variable

```
num = int(input())
```

```
cnt = 0
```

```
for i in range(2, num, 1):
```

```
    if (num % i == 0):
```

```
        cnt = cnt + 1
```

```
if (cnt == 0):
```

```
    print("prime number")
```

```
else:
```

```
    print("Not a prime number")
```

```
print(cnt)
```

<u>i</u>	<u>num % i</u>	<u>Result</u>	<u>cnt value</u>
1	5 % 1		
2	5 % 2 = 1	not divisible	0
3	5 % 3 = 2	not divisible	0
4	5 % 4 = 1	not divisible	0

cnt = 0 → prime number.

<u>i</u>	<u>num % i</u>	<u>Result</u>	<u>cnt value</u>
1			
2	4 % 2 = 0	divisible	cnt = cnt + 1 = 1
3	4 % 3 = 1	not divisible	cnt = 1

total cnt = 1 → not a prime number.

Flow Control statements - These are the statements that control how your program flows - that is, how & when different parts of the code are executed.

* There are 3 main types of flow control in Python.

<u>Type</u>	<u>Statement</u>	<u>Purpose</u>
1) Conditional	if, elif, else	Decision making
2) Looping	for, while	Repetition
3) Jump	break, continue, Pass	Alter loop behaviour

1) Break Statement -

Used to exit the loop immediately - it "breaks" the loop when a condition is met.

Ex -

```
for i in range(1,6):
    if i == 3:
        break
    print(i)
```

o/p =
1
2

Use when: you want to stop a loop early when once you've found what you need.

2) Continue Statement -

Skips the current iteration & jumps to the next one.

doesn't stop the whole loop.

Ex -

```
for i in range(1,6):
    if i == 3:
        continue
    print(i)
```

o/p =
1
2
4
5

3) Pass statement -

It is used as a placeholder when a statement is required syntactically, but you don't want any action yet.

Ex - for i in range(1,4):

if i == 2:

pass

print("Number:", i)

o/p =

Number : 1

Number : 2

Number : 3

Now will write prime program using break.

n1 = int(input("enter a num:"))

for i in range(2, n1, 1):

if (num % i == 0):

print("Its not a prime number")

break

else:

print("Its primenumber")

→ n1 = 6

i

n1 % i

is divisible?

o/p (Action)

2

6 % 2 = 0

Yes

print "not prime,"

break the loop.

Q) 1 1 1 1
 2 2 2 2
 3 3 3 3
 4 4 4 4
 5 5 5 5

⇒ For this we have to use nested loops.

Nested loops - for loop inside a for loop.

Syntax -

```
for i in range(): # outer loop
    for j in range(): # inner loop
        Statements of inner loop
    Statements of outer loop.
```

⇒ i → how many rows

j → how many elements in each row

In 1st line i=1, j should repeat 5 times.

We have to use end=" " for coming next to next in the same iteration.

Q) 1 1 1 1 1
 2 2 2 2 2
 3 3 3 3 3
 4 4 4 4 4
 5 5 5 5 5

```
Q) for i in range(1,6,1): o/p ⇒
    for j in range(1,6,1):
        print(i, end=" ")
    print()
```

1 1 1 1 1
 2 2 2 2 2
 3 3 3 3 3
 4 4 4 4 4
 5 5 5 5 5

```
Q) for i in range(1,6,1):
    for j in range(1,6,1):
        print(j, end=" ")
    print()
```

o/p ⇒ 1 2 3 4 5
 1 2 3 4 5
 1 2 3 4 5
 1 2 3 4 5
 1 2 3 4 5

Q) * i=1, j=1
 * * i=2, j=1,2
 * * * i=3, j=1,2,3
 * * * * i=4, j=1,2,3,4
 * * * * * i=5, j=1,2,3,4,5

```
for i in range(1,6,1):
    for j in range(0,i,1):
        print("*", end=" ")
    print()
```

o/p =>

```
*
* *
* * *
* * * *
* * * * *
```

Q) * * * * * i=5, j=1
 * * * *
 * * *
 * *
 *

```
for i in range(5,0,-1):
    for j in range(0,i,1):
        print("*", end=" ")
    print()
```

o/p =>

```
* * * * *
* * * *
* * *
* *
*
```

Q) 2 4
 2 4 6 8 10
 6 12
 4 8 12 16 20
 10 20

=> here, it's a table with even numbers odd numbers have space.

Suppose let's print

```
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
4 8 12 16 20
5 10 15 20 25
```

So, now same thing but only even, in odd space. <

```
for i in range(1,6,1):
    for j in range(1,6,1):
        if ((i*j)%2) == 0:
            print(i*j, end=" ")
        else:
            print(" ", end=" ")
    print()
```

for i in range(1,6,1):

for j in range(1,6,1):

print(i*j, end=" ")

print()